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EOR ATLAS

Historical EOR in Malaysian Fields

TOTAL OF FIELD	TOTAL OF EOR
14	9

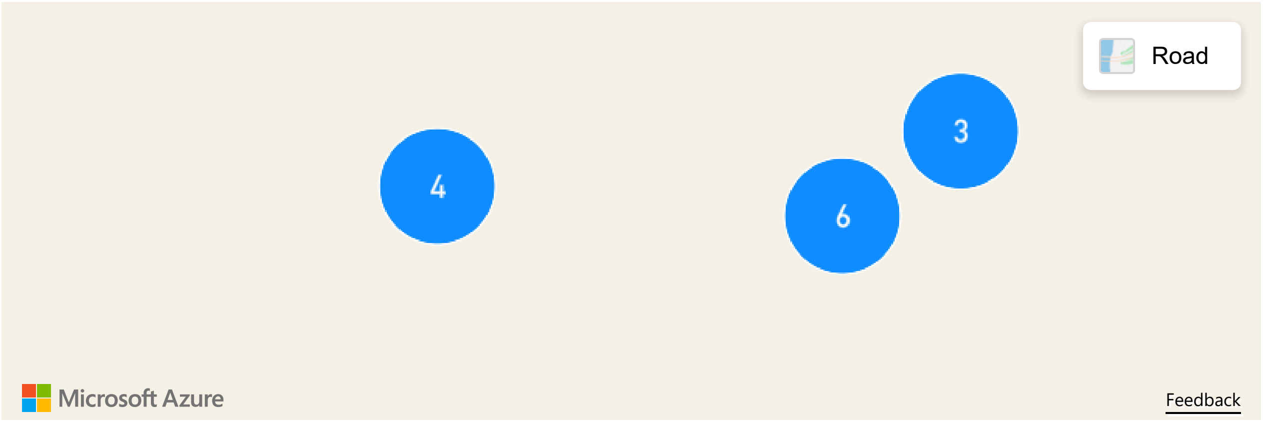
Types of E... : All

Field : All

Select your tool :-

- EOR Screening Tool
- Field Candidate EOR
- Past EOR Screening Result
- CEOR :Fluid - Fluid
- CEOR :Fluid - Rock
- Challenge + Lesson Learnt

Field ● Angsi ● Dulang ● Bokor ● Baronia ● Baram ● Guntong ● Samarang ● South Furiou... ● Tapis



Field	Type of Study	Types of EOR
⊞ Dulang	R&D /Lab Reports	CEOR Alkaline
⊞ Angsi	FDP Study	CEOR AS
⊞ Barton	TBD	CEOR ASP
⊞ South Furious 30	TBD	CEOR ASP
⊞ St Joseph	R&D /Lab Reports	CEOR ASP
⊞ Baronia	Evaluation Report	CEOR S-Foam
⊞ Baram	R&D /Lab Reports	EWAG
⊞ Bokor	R&D /Lab Reports	EWAG
⊞ Guntong	TBD	EWAG
⊞ Samarang	TBD	EWAG

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Enhanced Oil Recovery (EOR) Screening Tools

Clear Input

Past EOR Results

Reservoir Properties

Reservoir Type

Sandstone

Oil Gravity (API)

39.40

Oil Viscosity (cp)

1.000

Reservoir Depth (ft)

4271.00

Reservoir Pressure (psia)

2078.00

Reservoir Temperature (°C)

103

Average Permeability (md)

114.00

Porosity (%)

0.200

Initial Oil Saturation (%PV)

Water Chemistry

Salinity TDS (ppm)

Hardness Ca+Mg (ppm)

Gas Injection

Gas Availability

No | Yes

Oxygen in Present System

No | Yes

Volumetrics & RF

OOIP (MMstb)

111

Base Recovery Factor (%)

35

Drive Mechanism

Combination Drive

RESULTS FROM SCREENING INPUT

METHODS PASSING

0

METHODS FAILING

0

TOTAL METHOD

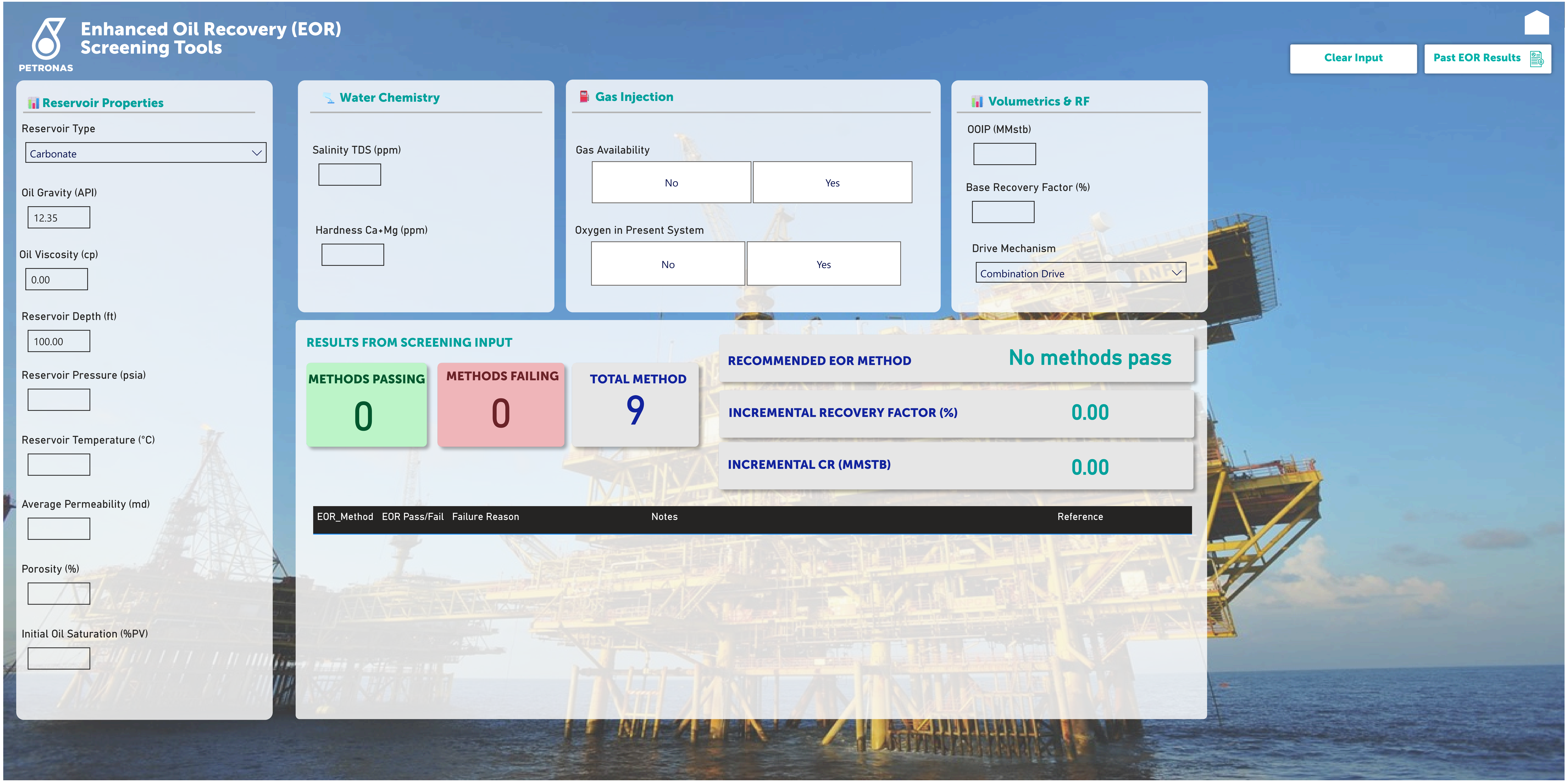
9

RECOMMENDED EOR METHOD: No methods pass

INCREMENTAL RECOVERY FACTOR (%): 16.50

INCREMENTAL CR (MMSTB): 56.84

EOR_Method	EOR Pass/Fail	Failure Reason	Notes	Reference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Enhanced Oil Recovery (EOR) Screening Tools
Clear Input
Past EOR Results
Reservoir Properties
Reservoir Type
Carbonate
Oil Gravity (API)
12.35
Oil Viscosity (cp)
0.00
Reservoir Depth (ft)
100.00
Reservoir Pressure (psia)
Reservoir Temperature (°C)
Average Permeability (md)
Porosity (%)
Initial Oil Saturation (%PV)
Water Chemistry
Salinity TDS (ppm)
Hardness Ca+Mg (ppm)
Gas Injection
Gas Availability
No
Yes
Oxygen in Present System
No
Yes
Volumetrics & RF
OOIP (MMstb)
Base Recovery Factor (%)
Drive Mechanism
Combination Drive
RESULTS FROM SCREENING INPUT
METHODS PASSING
0
METHODS FAILING
0
TOTAL METHOD
9
RECOMMENDED EOR METHOD
No methods pass
INCREMENTAL RECOVERY FACTOR (%)
0.00
INCREMENTAL CR (MMSTB)
0.00
EOR_Method
EOR Pass/Fail
Failure Reason
Notes
Reference

FIELD/RESERVOIR PARAMETERS

(CANDIDATES SCREENING)

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RESET ALL INPUT

TOTAL OF FIELDS
41

TOTAL OF RESERVOIRS
600

RF GAP x STOIIP (MMSTB)
1.31K

GRAPH PROPERTIES

X - Axis Parameter: Temp (deg C)

Y - Axis Parameter: EUR_ARPR 1.1.2026

INJECTED FLUID + PRODUCTION STATUS

INJECTED FLUID

Select all | GAS | NONE | WAT | WATGAS

RESERVOIR PRODUCING STATUS :

Select all | NONPRODUCING | PRODUCING

OIL / ROCK PROPERTIES

OIL API: 18.00 – 48.90

TEMPERATURE (C): 41.67 – 140.00

OIL COLUMNS (FT): 0.00 – 5,780.00

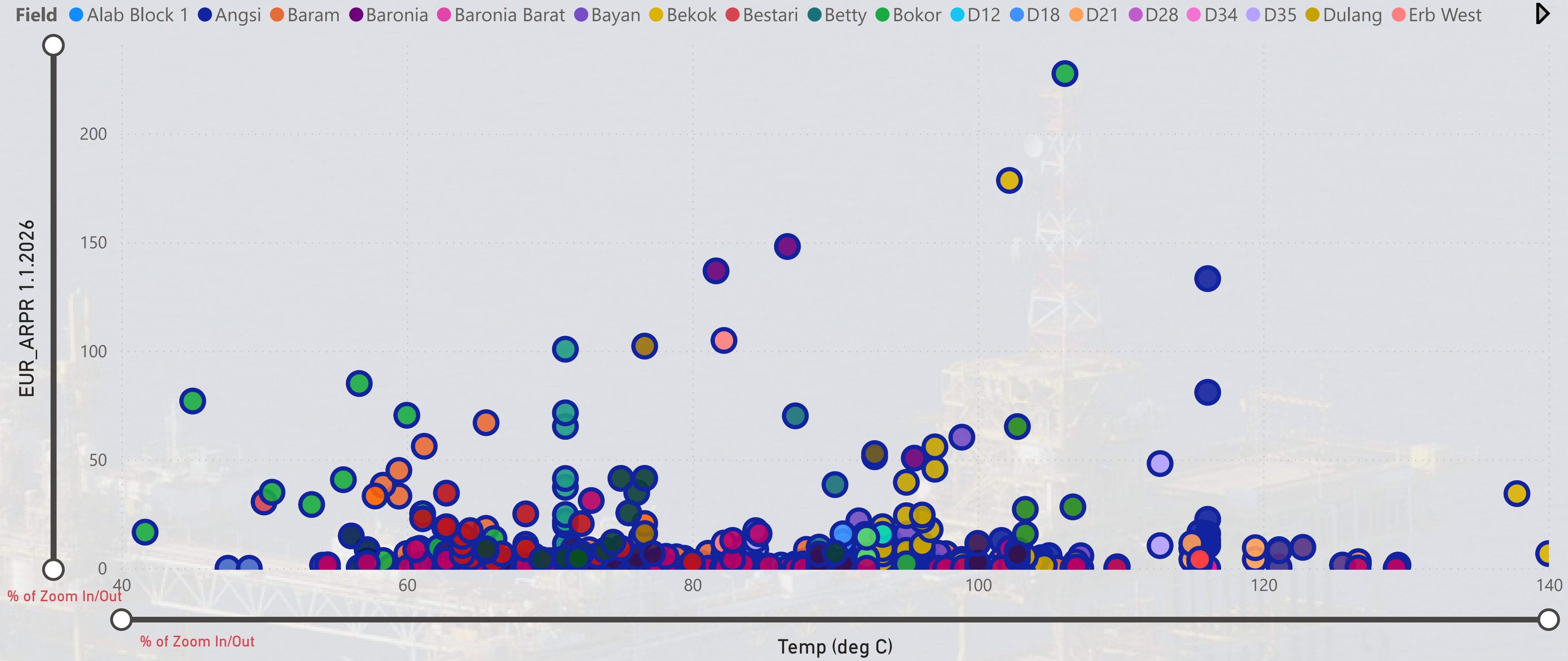
OIL VISCOSITY: 0.01 – 18.97

AVG POROSITY: 0.01 – 0.36

AVG PERMEABILITY: 0.01 – 6,600.00

FIELD CANDIDATES EOR

Z Axis (Categorical): Field



0 means no upside potential

Field	Temp (deg C)	Injected Fluid	Avg Permeability (mD)	Oil API	STOIIP_ARPR 1.1.2025	Sum of CR volume potential = RF Gap x STOIIP (MMSTB)	Sum of Oil Producers
⊞ Alab Block 1	142.22	NONE	3,600.00	58.00	7.91	1.15	3
⊞ Angsi	1,701.11	NONE	8,020.00	630.50	827.48	27.11	88
⊞ Baram	10,913.89	NONE	37,887.57	4,672.64	1,336.93	148.03	494
⊞ Baronia	263.89	WATGAS	495.00	126.00	752.02	12.50	89
⊞ Baronia Barat	190.00	NONE	20.00	66.00	25.57	0.20	2
⊞ Bayan	1,548.33	GAS	9,095.00	592.00	505.75	44.22	83
⊞ Bekok	581.11	GAS	470.00	237.10	667.86	44.90	92
⊞ Bestari	50.00	WAT	230.00	28.00	123.10	18.97	6

CEOR
Fluid - Fluid Interaction

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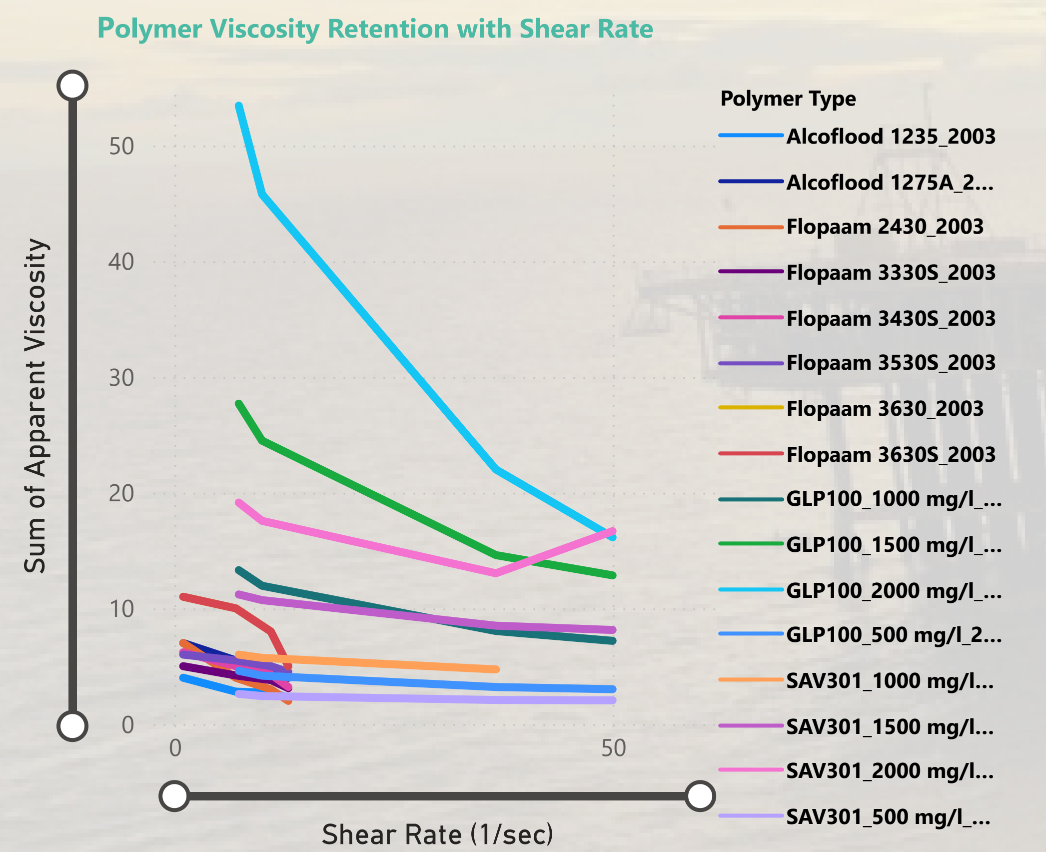
RHEOLOGY

YEAR, TEMP & POLYMER TYPE

- Select all
- 2003
- 2013
- 2025
  - 80
    - SuperPusher SAV 82 XV_2025
  - 90
    - Flopaam AN 132 VHM_2025
    - SuperPusher SAV 73 XV_2025
  - 120
    - SuperPusher SAV 10_2025

KEY OBSERVATIONS :

- Thermal Stability : Polymer studied in 2025 (SNF) has longer viscosity retention at higher temperature (80-150 deg C) compared to polymers studied in 2013 & 2003
- 2013 ANGSI CEOR FDP : 2 polymers proposed are Super Pusher SAV 301 & GLP100 (relative best among polymers studied in 2013)

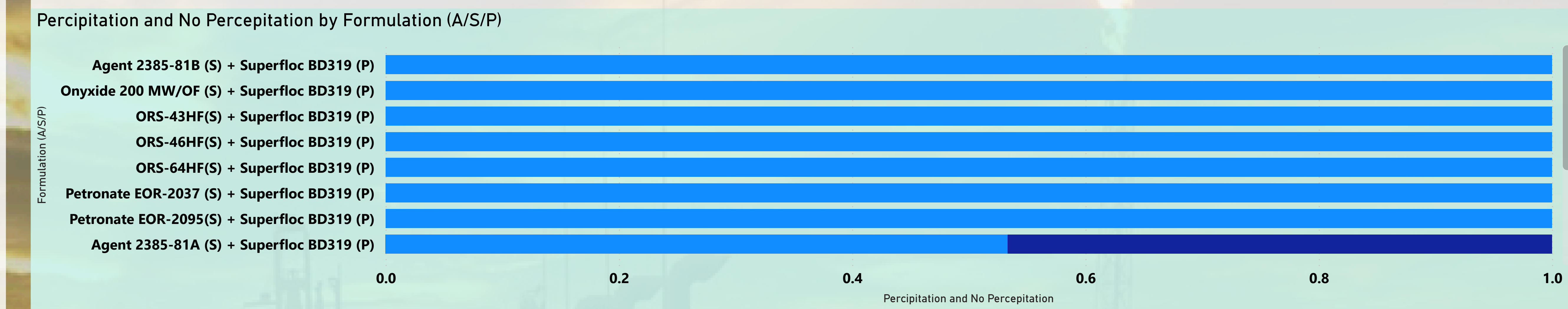


- Select all
- 2003
- 2013
  - GLP100_1000 mg/l_2013
  - GLP100_1500 mg/l_2013
  - GLP100_2000 mg/l_2013
  - GLP100_500 mg/l_2013
  - SAV301_1000 mg/l_2013
  - SAV301_1500 mg/l_2013
  - SAV301_2000 mg/l_2013
  - SAV301_500 mg/l_2013

KEY OBSERVATIONS :

- 2013 polymer showed better polymer viscosity retention at higher shear rate compared to polymer studied in 2003

PHASE BEHAVIOUR TEST



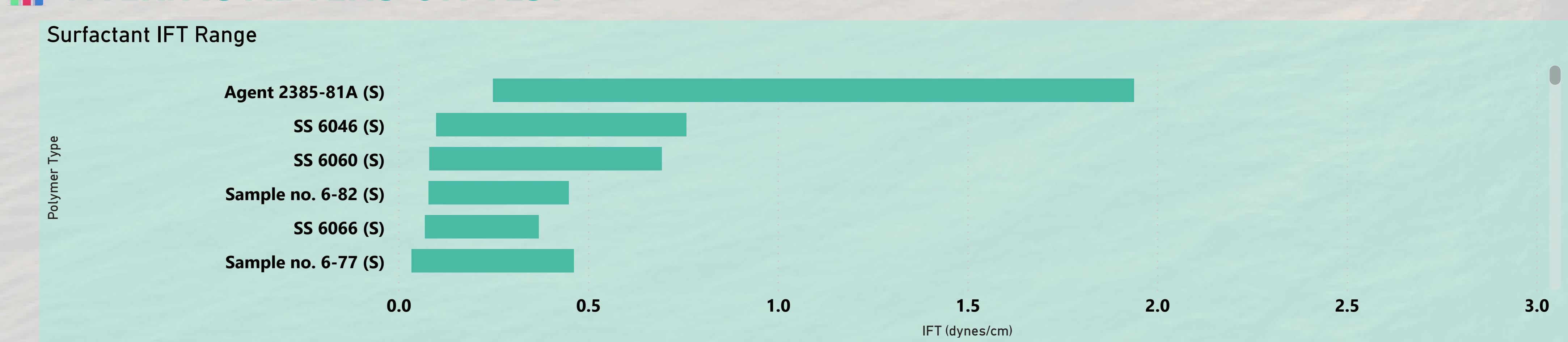
Out of 14, only 7 surfactants passed the Phase Behaviour Test with no total Precipitation at various brine and surfactant concentration

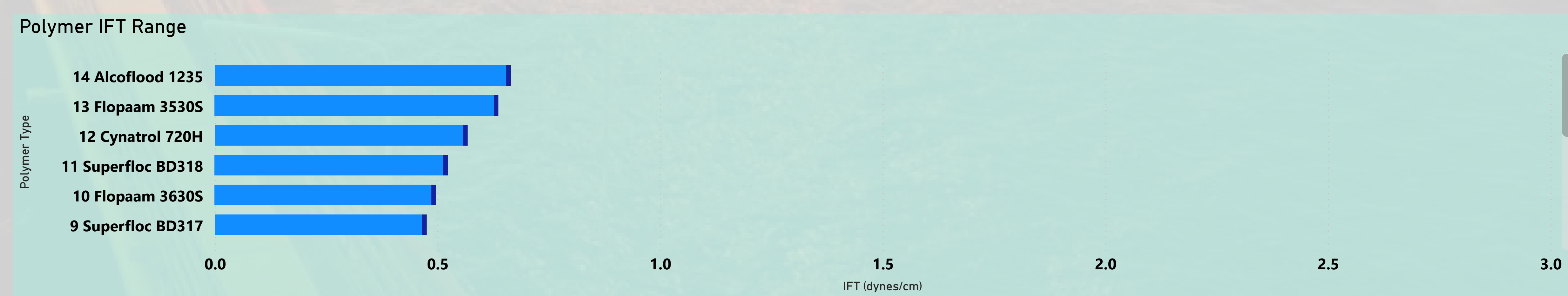
- Na_2CO_3 alkali is more stable compared to NaOH with selected surfactant
- FDP 2014 revised from NaOH to Na_2CO_3 compared to FDP 2011

Surfactant Formulation 43 passed the Phase Behaviour Test

To proceed to CoreFLood

INTERFACIAL TENSION TEST





KEY OBSERVATIONS :

- In 2003: Out of 14 surfactants studied , only 7 surfactants passed PB test underwent IFT test. 1 showed favorable IFT ,the lowest , Sample 6-79 (0.003-0.97 dynes/cm)
- In 2013: Out of 43 combinations of surfactants studied , only Formulation 43 which passed PB test underwent IFT test with IFT of 0.0008-0.13 dynes/cm.

Surfactant by Year & PB Test Results

- Select all
- 2003
  - No
  - Yes
- 2013

KEY OBSERVATIONS :

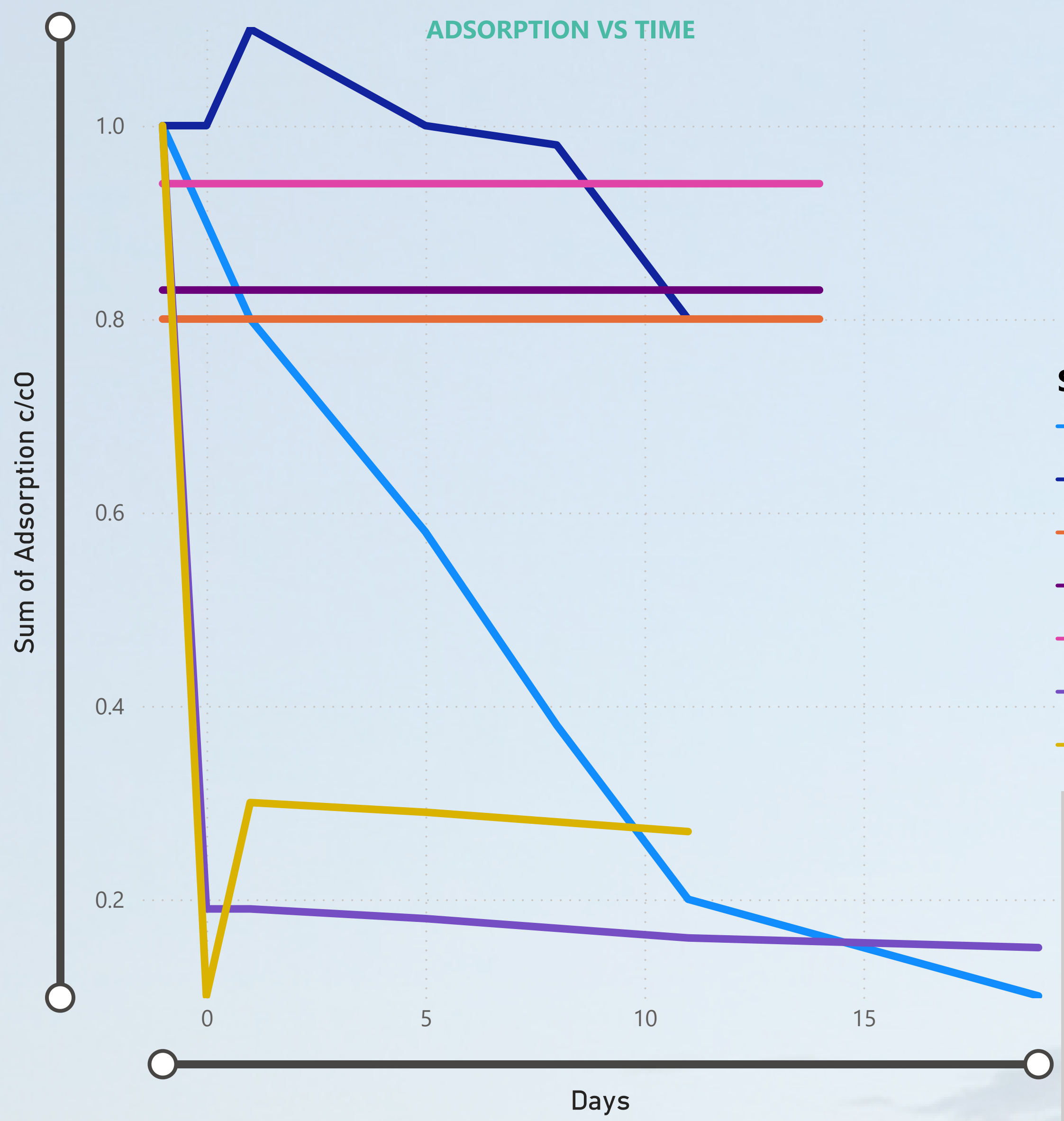
- In 2003: No polymer IFT test was conducted
- In 2003: Out of 14 surfactants studied , only Formulation 43 surfactants passed PB test underwent IFT test with ~ 0.359 dynes/cm)

Polymer (Year)

- Select all
- 2013

CEOR
Fluid - Rock Interaction

ADSORPTION TEST



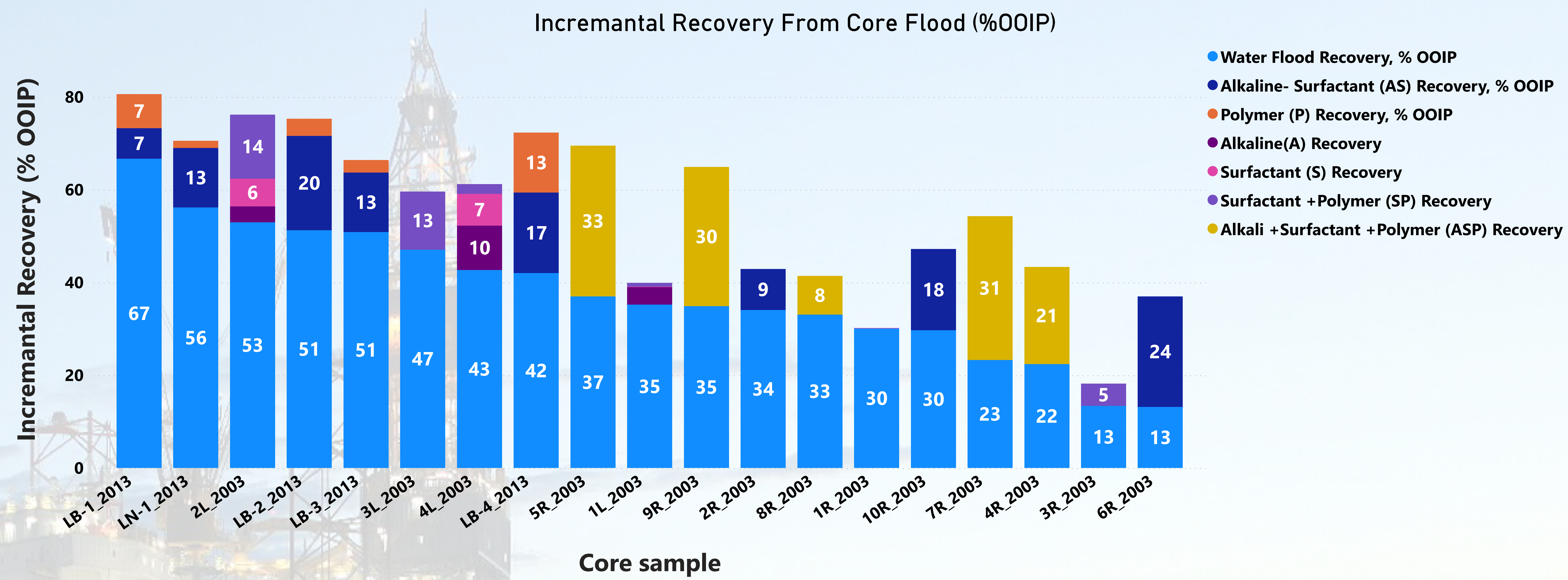
YEAR, TEMP & POLYMER TYPE

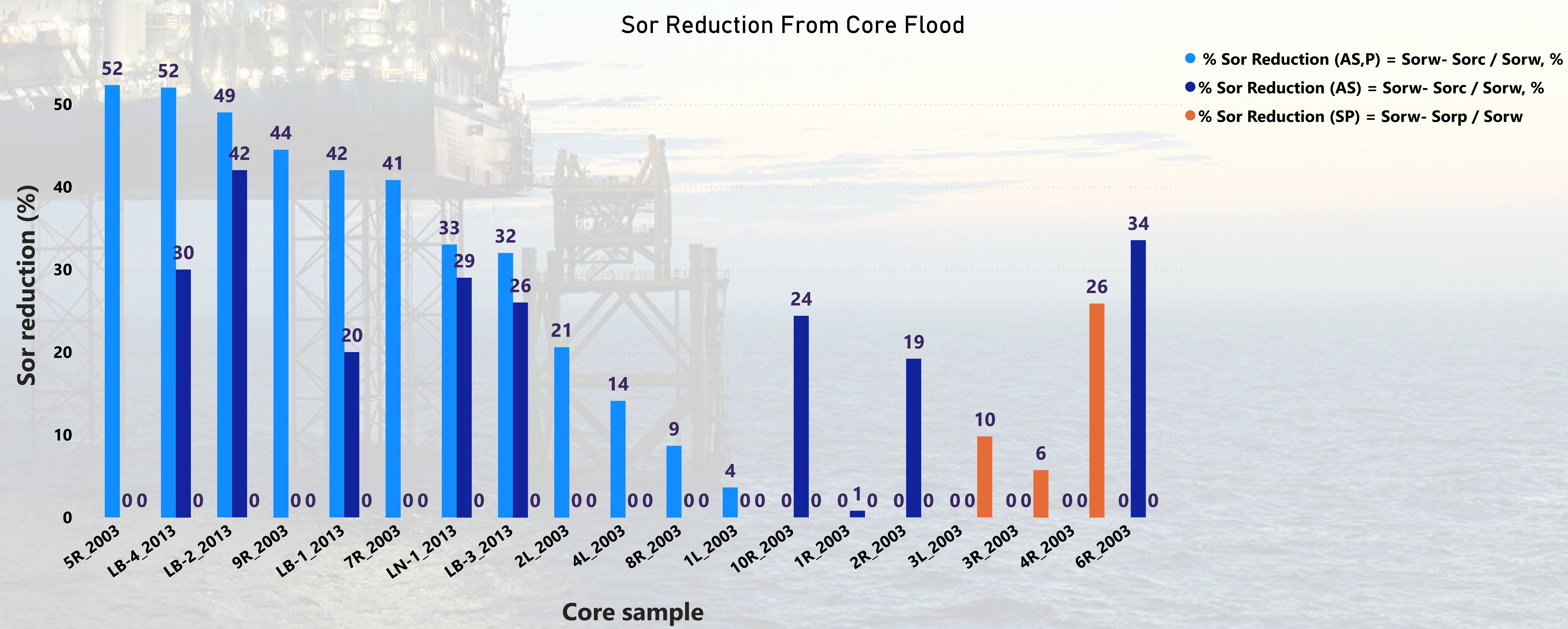
- Select all
- 2003
  - Angsi
    - Alcoflood 1235_(no alkali)_...
    - Alcoflood 1235_(with NaoH...
    - SS-7064_(no alkali)_2003
    - SS-7064_(with NaoH)_2003
- 2013

KEY OBSERVATIONS

- **2003** : Surfactant and polymer loss was more pronounced for solutions made in seawater (no alkali) . **Alkali is important in reducing surfactan adsorption loss.**
- **2013** : Only **Formulation 43 surfactant** underwent static adsorption test with 0.8-0.94 mg/g adsorption range.

CORE FLOOD





CORE FLOOD

Year	Core Floof Sample	Type	Alkali Type & Concentration	Surfactant Type & Concentration	Polymer Type & Concentration
2003	10R_2003	AS	1 % NaOH	0.075% 6-72	NA
2003	1L_2003	ASP	% NaOH	ORS 6-49	Flopaam 3630ST
2003	1R_2003	S	NA	0.1% 6-79	NA
2003	2L_2003	ASP	% Na2C03	6 -72	Flopaam 3430ST
2003	2R_2003	AS	0.05% NaOH	0.05% 6-72	NA
2003	3L_2003	SP	NA	6 -79	HE 300
2003	3R_2003	SP	NA	0.15% 6-79	0.05% HE300
2003	4L_2003	ASP	NA	6 -72	AN125
2003	4R_2003	ASP	1 % NaOH	0.075% 6-72	0.03% AN125
2003	5R_2003	ASP	1 % NaOH	0.075% 6-72	0.05% AN125
2003	6R_2003	AS	1 % NaOH	0.075% 6-72	NA
2003	7R_2003	ASP	1.25 % NaOH	0.1% 6-72	0.03% AN125
2003	8R_2003	AS+P	1 % NA2Co3	0.05% ORS-164	0.03% AN125
2003	9R_2003	ASP	1 % NaOH	0.075% 6-72	0.03% AN125
2013	LB-1_2013	ASP	1 % NaOH	0.075% ORS6-72	1000 ppm SuperPusher
2013	LB-2_2013	ASP	1 % Na2C03	0.075% ORS6-72	1000 ppm SuperPusher
2013	LB-3_2013	ASP	1 % NaOH	0.075% ORS6-72	1000 ppm SuperPusher
2013	LB-4_2013	ASP	1 % Na2C03	0.075% ORS6-72	1000 ppm SuperPusher
2013	LN-1_2013	ASP	1 % NaOH	0.075% ORS6-72	1000 ppm SuperPusher

EOR ATLAS

Past EOR Screening Results

PAST EOR SCREENING INPUT

- [ ] Select all
- [ ] ALSP
  - [ ] Barton
    - [ ] G/H
- [ ] CO2 WAG
  - [ ] Tiong
    - [ ] J-20/21
- [ ] CO2M NON WAG
  - [ ] Kepong
- [ ] CO2M WAG
  - [ ] East Bunga Kekwa
  - [ ] East Bunga Orchid
  - [ ] Guntong
  - [ ] Tabu
  - [ ] Tapis
  - [ ] Tinggi
  - [ ] West Bunga Kekwa

SOURCE :

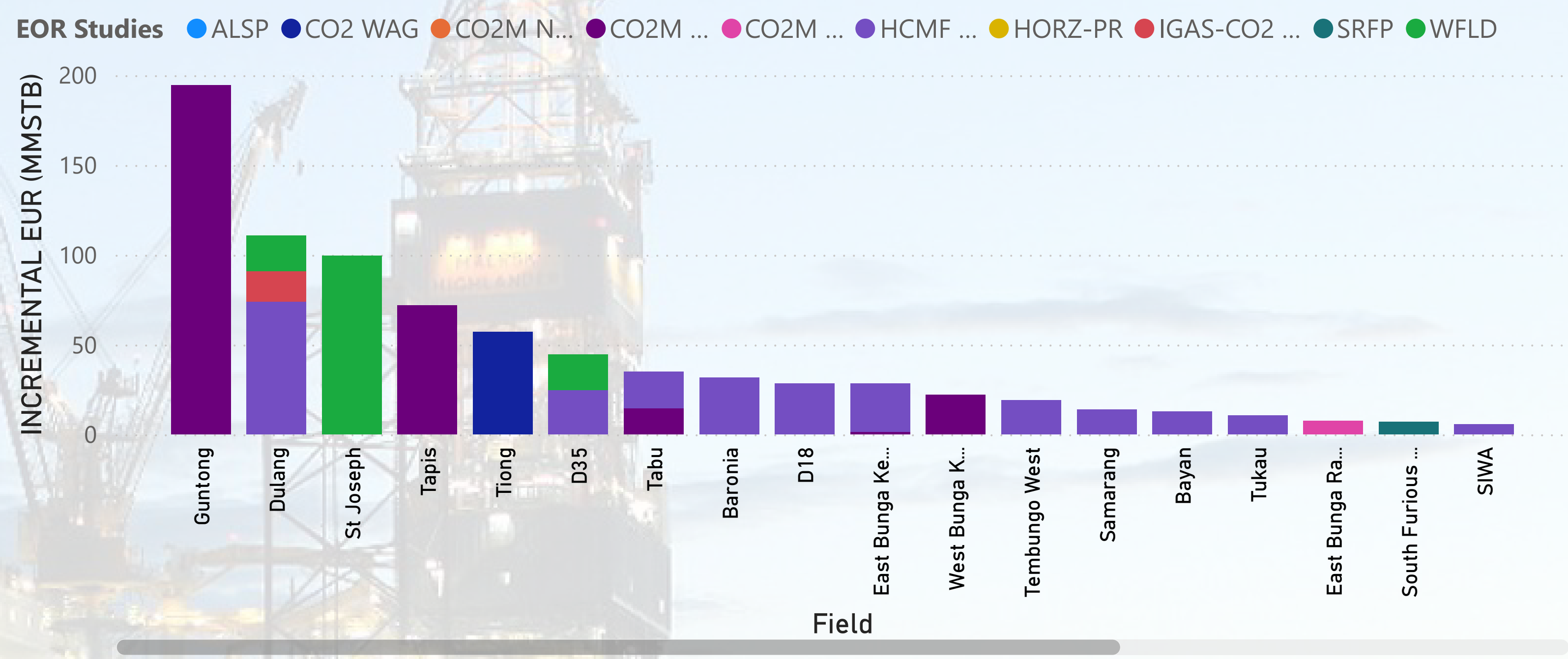
TITLE: YEAR 2000 IOR/EOR SCREENING FOR MALAYSIA'S OIL PRODUCING FIELDS

YEAR: 2001

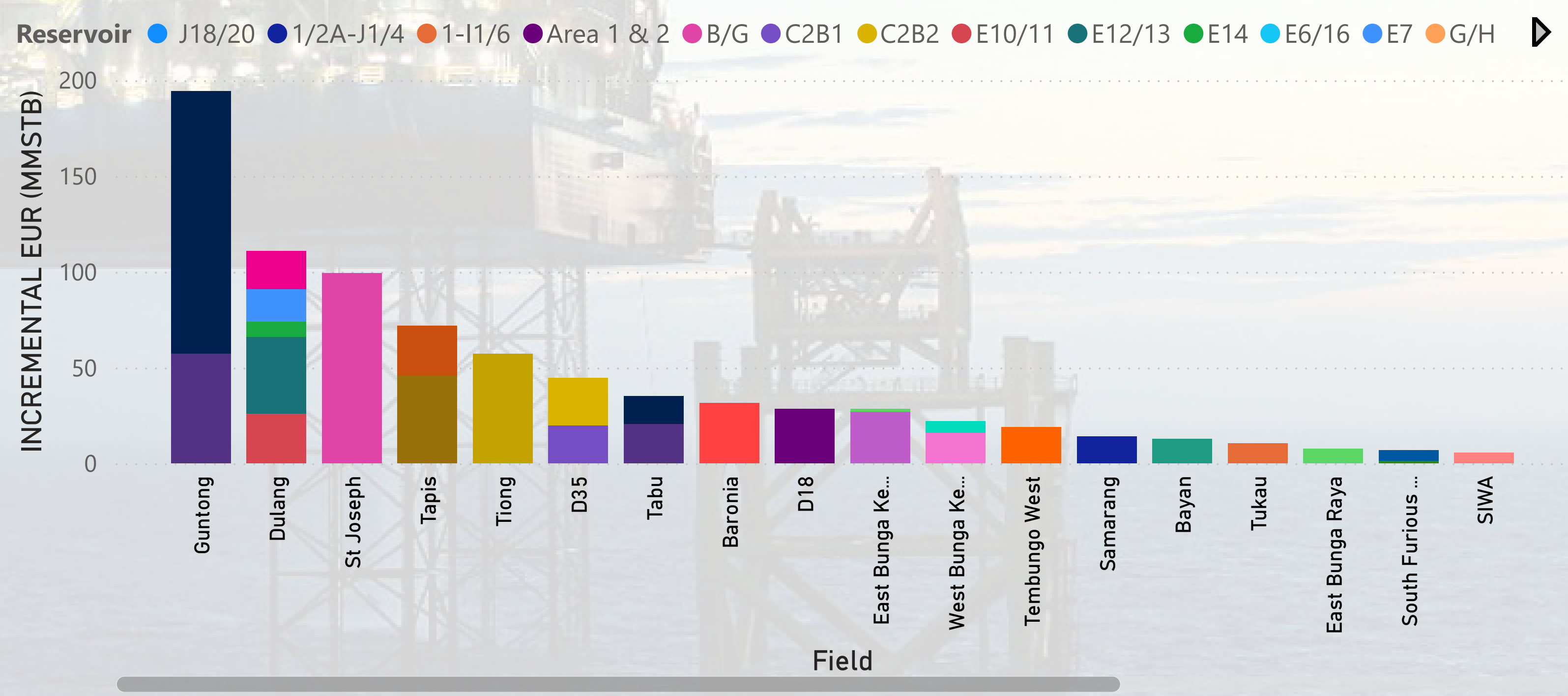
ACCESS PAPER HERE

PAST EOR SCREENING RESULT

INCREMENTAL EUR (MMSTB) by Field and EOR Studies



INCREMENTAL EUR (MMSTB) by Field and Reservoir



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EOR ATLAS

Challenges + Lesson Learnt

Field

All

FIELD

Field ● Angsi ● Dulang ● Bokor ● Baronia ● Baram ● Guntong

CHALLENGES AND LESSON LEARNT

Field	Area	Challenges	Lessons
Angsi	Chemical Compatibility & Water Treatment.	Sensitivity of alkaline formulations to divalent ions and added complexity of offshore water softening; desalination increased weight and operational load.	Offshore i compatibi units sign weight.
Angsi	Chemical Cost & Logistics	High cost and large storage volume required offshore	Offshore c major cos
Angsi	Chemical Validation	Lab results needed confirmation in field conditions	Laborator guarantee
Angsi	Facilities & Infrastructure	CEOR vessel design exceeded weight expectations	Early desi EOR facilit
Angsi	Fluid Handling & Pumping Systems	High-concentration injected fluids caused elevated pressure and pumping difficulties; handling mixed oil-water-gas phases offshore added complexity.	Reliable a managem CEOR ope
Angsi	Simulation & Risk	Uncertainty in Sor, adsorption, and IFT performance	Static and